Assignment 4 test plan: overtaking, state machine, stop-sign camera pan

Test13 build (UNO_Mapping + ESP32_Assignment2, UV 43). Tester: _____ Car: _____ Date: _____

Safety: wheels off the ground for the first pass of each stage, hand on STOP CAR, obstacle = soft (the moose toy). Overtaking needs a clear straight: about 125 cm between the detection point and the next corner or the intersection (best place: the leg after the intersection, O to D, or the first part of C to O). Pass criteria are expectations; tick a box when the step passes.

Stage 0 - Upload and memory

	Step	Expected / pass	Result / notes
☐	Upload both sketches (UNO switch OFF). Write down the IDE flash % and RAM %.	Flash under 100%, RAM under 75%. If not: in MapConfig.h keep FOLLOW_ENABLED 0, then PAN_SERVO / IMU_QMI8658 per README.	
☐	Open the page; read the title and status pill.	Test13, no "UNO version mismatch".	
☐	Check /traffic-status: free RAM FR.	Note the value; must not shrink while driving.	

Stage 1 - State machine log (wheels up)

	Step	Expected / pass	Result / notes
☐	Start a run, then press STOP. Open the Map tab.	A state timeline and a state-transition table appear (TRACK, CORNER, SEARCH, STOP SIGN, WAIT GREEN, OVERTAKE ..., FAULT, FINISHED).	
☐	Show a red card, then a green card.	RED CHECK / WAIT GREEN appear and end in the log with times.	
☐	Download the transitions CSV.	One row per transition: time, from, to, x, y, heading.	

Stage 2 - Camera pan toward the STOP sign

	Step	Expected / pass	Result / notes
☐	Use the Camera pan test buttons (left / centre / right). Find the angle where the STOP sign fills the image from the approach.	Write the angle: _____ (and which direction is right: _____)	
☐	Set the pan angle and "start looking X cm before the intersection". Start a run.	Camera turns toward the sign only near the intersection (state APPROACH), then returns to centre after it.	
☐	Run 5 passes at the intersection.	STOP sign recognised and 3 s hold each pass: ___ / 5. Without the pan (angle 0) for comparison: ___ / 5.	
☐	Watch the ultrasonic while the camera is turned.	No E-stop caused by the sign or the wall; obstacle detection resumes about 0.4 s after the camera is straight again.	

	Step	Expected / pass	Result / notes
☐	Slow-down percentage in the approach window.	Car slows near the intersection, resumes normal speed after crossing.	

Stage 3 - Corner decisions and map auto-correction

	Step	Expected / pass	Result / notes
☐	Run 3 laps. Open the segment table in the Map tab.	Every corner shows expected turn (L/R) and actual turn; no "wrong way".	
☐	Check the intersection resync (summary).	Crossing detected twice per lap; progress snaps to the known 150 cm (C to O) / 120 cm (E to O).	
☐	If the car turns the wrong way at a corner: note the corner and Download all lines.	Send me the file and the corner name.	

Stage 4 - Overtaking (static obstacle)

	Step	Expected / pass	Result / notes
	Overtaking options on the Drive tab: ON, side = Auto (outside of the next turn)		
☐	Put the moose on the line on the leg after the intersection (about 25-45 cm after O). Start a run.	Car stops 30-35 cm before it, steps OUT 90 degrees, drives alongside, steps back IN, finds the line, continues. States OVERTAKE OUT / PASS / RETURN logged.	
☐	Which side did it pass?	Outside of the upcoming turn (left on C-O-D). Not on the inside: _____	
☐	Tune: side clearance, pass length (obstacle length + clearance), join advance (distance wheel axle to line sensors).	Record: clearance ___ cm, pass ___ cm, join ___ cm. No contact with the moose.	
☐	Repeat 10 times.	Successful overtakes: ___ / 10 (touches: ___, faults: ___)	
☐	Put the moose where there is no room (near a corner or inside the intersection).	Car waits behind it (OVERTAKE WAIT) or E-stops; never touches it. Move the moose away and it continues.	
☐	Different obstacle shapes and sizes (box, bottle, larger).	Record success for each; widen clearance if needed.	

Stage 5 - Moving robot ahead

	Step	Expected / pass	Result / notes
☐	Another robot on the same line, slower, same direction. Set the pass length longer (pass length x speed / speed difference).	Overtakes on the clear straight on the outside, resumes tracking. If there is no room it waits behind until room appears.	
☐	Overtake attempts: ___ successful: ___ rate: ___ %	Needed for the report (success rate of the Bug algorithm).	

Stage 6 - Traffic rules with an obstacle

	Step	Expected / pass	Result / notes
☐	Red light appears while following or overtaking.	Car stops (no violation), continues on green.	
☐	STOP sign at the intersection: 3 s hold every time, also in laps with an obstacle.	Hold >= 3 s, no violation.	
☐	Count: traffic light violations ___, stop sign violations ___, off-road > 3 s ___	Score sheet items; the page lists search episodes longer than 3 s.	

Stage 7 - Score sheet data (per heat)

	Step	Expected / pass	Result / notes
☐	After each heat read the Map summary.	Completion time, average speed, lap times, overtaking time, overtaking success.	
☐	Check the map.	Stop sign, traffic light, intersection and obstacle positions are drawn in the inertial frame; overtaking trajectory is highlighted.	
☐	Save: map PNG, heading chart, state timeline, transitions CSV, points CSV, Download all lines.	Heat 1 / 2 / 3 saved: ___ / ___ / ___	

Stage 8 - Regression

	Step	Expected / pass	Result / notes
☐	Two clean laps with overtaking OFF.	As Test11: no change in behaviour.	
☐	E-stop (web, obstacle), gyro readout, zero button, wait for green, laps limit.	All work.	

If anything fails

Send "Download all lines" and the Summary text, plus the stage, step, and what the car did.